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Subcutaneous Ports for Chronic Nerve Cuff and Intramuscular Electrode Stimulation in Animal Models

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Abstract

Objective. Tracking recovery after nerve injury may require many intermittent assessments over long periods, preferably with non- or minimally invasive methods. We developed subcutaneous electrical connection ports (ECPs) for repeated connection to nerve cuff or intramuscular electrodes via transdermal needles and evaluated them during studies of laryngeal reinnervation.

Study Design. Animal experiment.

Setting. Laboratory.

Methods. ECPs were designed and 3-dimensionally printed for connection to bipolar electrodes with biocompatible polymers. Dual compartments filled with conductive silicone capped with nonconductive silicone were used to make the connections between electrode leads and transdermally inserted needles. Ten dogs (19-29 kg) were implanted with 22 ECPs. In 7 dogs, 11 electrodes were placed on recurrent laryngeal nerves proximal to transection and suture repair to track laryngeal reinnervation. In 6 dogs, 8 spinal accessory nerve cuff electrodes were used to stimulate neck muscle contraction. In 2 dogs, 3 electrodes were implanted in the thyroarytenoid muscle. Stimulation thresholds, electromyography, and videolaryngoscopic imaging were obtained in 156 tests over survival periods up to 32 months. Stimulation data provided information about ECP performance.

Results. ECPs added negligible resistance to electrodes (mean $\pm$ SD, 2.14 $\pm$ 0.9 Ω). Despite some electrode leads breaking distally, ECPs were reliable and well tolerated at implant sites and enabled periodic assessment of nerve and muscle function over the time course of laryngeal reinnervation. Histology showed ECP encapsulation as thin layers of connective tissue and minimal acute inflammation.

Conclusion. Custom ECPs are easily fabricated and cause little tissue reaction over months to years of subcutaneous implantation, facilitating long-term physiologic studies.

Keywords

intramuscular electrode, nerve cuff electrode, nerve stimulation, recurrent laryngeal nerve regeneration

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Longitudinal studies of nerve regeneration or functional electrical stimulation often involve periodic recording or stimulation through chronically implanted nerve cuff and intramuscular electrodes. Direct electrical connection with implanted electrodes can be achieved by having wires protrude through the skin or by attachment to a transcutaneous socket.[1] Unfortunately, these approaches present a substantial risk of infection—particularly across survival periods that can span months to years (for review see Navarro et al[2]).

An alternative, inexpensive, and minimally invasive approach to access implanted electrodes and sensors is to use a ''data port'' or ''data block.''[3-5] Laboratory apparatus can be attached to an implanted electrode by using an electrically insulated needle with an uninsulated tip and by inserting it through the skin and the self-sealing surface of an implanted port attached to the chronically implanted electrode or sensor. The needle tip will then make electrical contact with the electrode lead(s) through an intermediate conductive filler within the port's compartment.

Data ports in the past have been machined from implant-grade materials, but modern access to 3-dimensional printers with biocompatible photopolymers allows researchers to readily design and fabricate custom ports to suit their experimental needs. Our goal was to develop easily fabricated and/or modified electrical connection ports (ECPs) to allow minimally invasive electrical connection with chronically implanted nerve cuff or intramuscular electrodes for use in canine models of laryngeal paralysis and functional electrical stimulation.[6]

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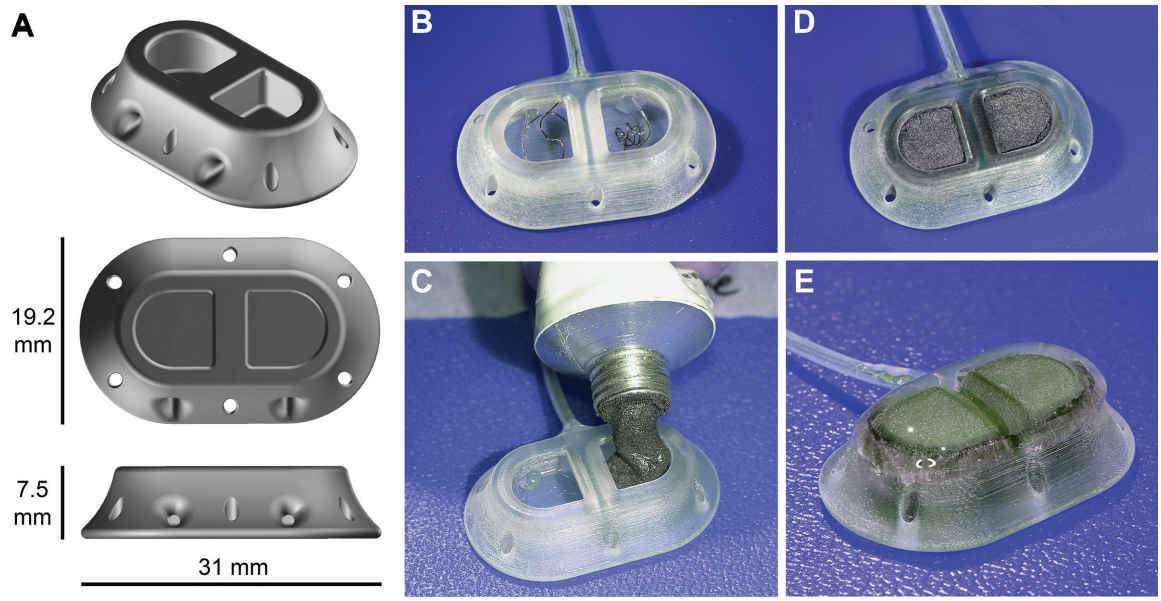


Figure 1. (A, B) The ECP was designed with SolidWorks software and 3-dimensionally printed with photopolymer. Electrode wires in ECP compartments are (C) covered in conductive silicone, which (D) cures and (E) is coated with nonconductive silicone. ECP, electrical connection port.

Methods

Port Materials and Design

ECPs were initially made with 40-mm lengths of thick-walled, platinum-cured silicone tubing with an inner diameter of 3.2 mm and outer diameter of 6.4 mm (9580-05; Cole-Parmer). The tubing provided a needle-penetrable body with a hollow compartment that could be filled with conductive, room temperature vulcanizing silicone. The tubing design worked well (functioning 2.6 years in 1 animal; see Results), but we sought a solid substrate for the compartment that would stop the inserted needle within the conductive silicone. Five bipolar electrodes were made with 10 PEEK fittings (polyetheretherketone, 5063-6591; Agilent) measuring 12 × 7 × 9 mm with a 0.5-mL compartment, and 2 bipolar electrodes were connected to 4 polysulfide ports (ClearPort-P/Medium; Access Technologies) with a 23-mm diameter, 10-mm height, and 0.36-mL compartment. These designs worked well but introduced more implant mass than necessary because each compartment had an independent enclosure.

Our final ECP design resembled the general shape and size of the medium-sized Access port used in animal research but with an oblong shape (31 × 19.2 × 7.5 mm) designed with SolidWorks software (Dassault Systèmes SE) that provided 2 independent port compartments per ECP (**Figure 1A**). Each compartment was 0.5 mL with a 1.5-mm-thick floor and 0.9-mm-thick walls. The design was printed with a rigid, semitransparent, biocompatible photopolymer (MED610; Stratasys Ltd) and Polyjet printer (Objet 260V; Stratasys Ltd; **Figure 1B-E**). ECPs were connected to electrode wires by removing approximately 20 to 40 mm of the electrode wire insulation, inserting the bare wire into the compartment, and adhering it at the entry point with nonconductive, implant-grade, room temperature vulcanizing silicone (40064; NuSil Technology LLC; **Figure 1B**). The ECP compartments were filled with nickel graphite conductive silicone (MMS-020-1; Moreau Marketing and Sales; **Figure 1C**) and cured overnight at 60 °C. The compartment opening was then covered with 2 to 3 mm of nonconductive silicone (**Figure 1E**) and oven cured for an additional 1 to 2 days. The conductive silicone remained needle penetrable after curing, and the overlying nonconductive silicone layer adequately sealed after needle electrode penetration and withdrawal. The dual-compartment printed ECPs were used with 14 chronically implanted bipolar electrodes (28 compartments).

ECPs were connected to bipolar nerve cuff electrodes made by MicroProbes Life Sciences Inc or custom-fabricated intramuscular electrodes made from 30 AWG silicone-jacketed stranded stainless-steel wire (AS814; Cooner Wire Inc; **Figure 1**). The intramuscular electrodes each had two 1- to 2-mm-wide exposed contacts wrapping circumferentially around silicone tubing (outer diameter, 2-4 mm) and spaced approximately 5 to 8 mm apart (**Figure 2**). Longitudinal electrodes have been used in canine vocal fold stimulation studies,[6,7] implanted parallel to the thyroarytenoid (TA) muscle fascicles, as done in the present study (see Figure 1 of Ludlow et al[7] and Figure 4 of Heaton et al[6]).

The nerve cuff electrodes were 6-mm silicone tubes (split longitudinally) with 1- to 1.5-mm internal diameters, 100-μm platinum wire ring contacts spaced 2 mm apart, and 3 embedded tie-down sutures (**Figure 3**). Wire leads from the electrodes to the ECPs were 60 to 100 mm in length. All electrodes and finished assemblies were washed and sterilized before implantation. ECPs made of PEEK or polysulfone were autoclaved, whereas ECPs printed with photopolymer were gas sterilized with ethylene oxide.

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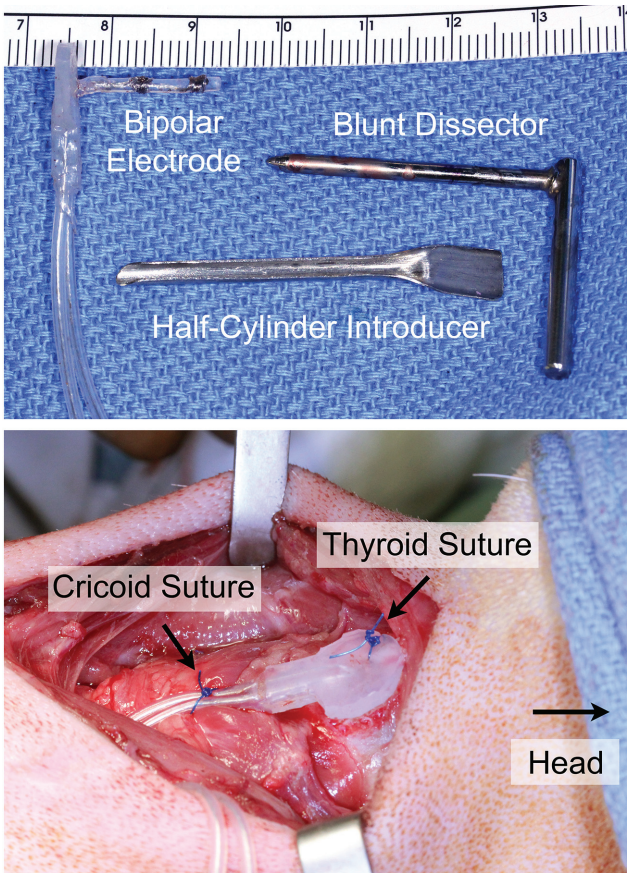


Figure 2. A bipolar intramuscular electrode is shown with dissector and introducer tools (top) and after being implanted into the left vocal fold and sutured to the cricoid and thyroid cartilages (bottom).

Bench-Top ECP Testing

Direct current resistance was measured between a 20G stainless-steel Huber needle inserted into each ECP compartment and electrode contacts with a multimeter to determine resistance added by the ECP for each fabricated electrode. Potential current leakage due to repeated needle penetration into the ECP compartment was tested by measuring alternating current impedance (at 1 kHz) between a monopolar needle electrode (019-475100; Natus Medical Inc) and 0.9% saline solution in which the ECP was submerged (ie, modeling an implanted electrode). Possible current leakage was measured across a series of 200 insertions into 2 ECP conductive compartments to represent chronic testing.

Implantation Surgery

Ten female dogs (19-29 kg) were chronically implanted with 22 bipolar electrodes during surgery under intravenous propofol anesthesia (0.3-0.5 mg/kg/min). In 7 dogs, 11 bipolar nerve cuff electrodes were placed on 1 or both recurrent laryngeal nerves (RLNs) at the time of nerve injury, proximal to transection and epineurial suture repair to track laryngeal reinnervation. In 6 dogs, 8 bipolar nerve cuff electrodes were placed on a branch of the spinal accessory nerve to stimulate cleidocephalicus (CC) muscle contraction without experimental nerve injury. In 2 dogs, 3 bipolar intramuscular electrodes were implanted in the TA muscle through 3.5-mm biopsy punch holes made in the thyroid lamina. A path along the long axis of the vocal fold was made for the electrodes with a blunt dissecting tool, and electrode insertion was assisted with a half-cylinder introducer tool (**Figure 2**).

After electrode implantation, a tunnel was created below the skin to provide a location for the ECPs distant from incisions. ECP versions with suture holes were then sutured to the skin surface with 2 to 4 sutures (3-0 Prolene) to prevent migration and were removed after 2 to 4 weeks. Representative surgical implantation of ECPs with cuff electrodes is shown in **Figure 4**. Experiments were performed in accordance with the US National Research Council's *Guide for the Care and Use of Laboratory Animals* and under protocols approved by the Pine Acres Rabbitry Farm Animal Care and Use Committee.

Implanted Electrode Testing Through ECPs

For periodic electrode testing across the postimplantation survival period, dogs were anesthetized as described and placed in suspension for direct laryngoscopy, viewed through a surgical microscope (M525 F50; Leica). Monopolar needle electrodes were temporarily inserted into the ECPs through the neck skin in 156 postimplantation electrode tests, spanning survival periods up to 32 months. Nerve stimulation via this percutaneous connection evoked compound muscle action potentials (CMAPs) and allowed measurement of neuromuscular stimulation thresholds causing visible neck or vocal fold movements. The location and orientation of ECPs were palpable and unambiguous under the skin

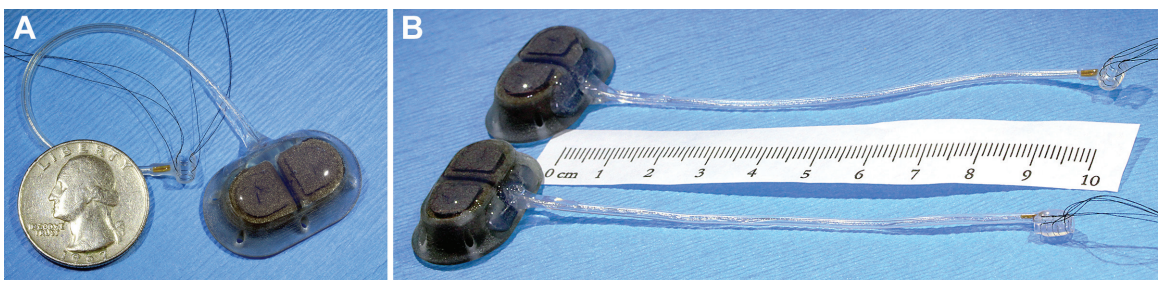


Figure 3. Bipolar nerve cuff electrodes with 1.5-mm internal cuff diameters and embedded silk sutures for cuff closure are shown attached to electrical connection ports. (A, B) Photos demonstrate scale and length in centimeters.

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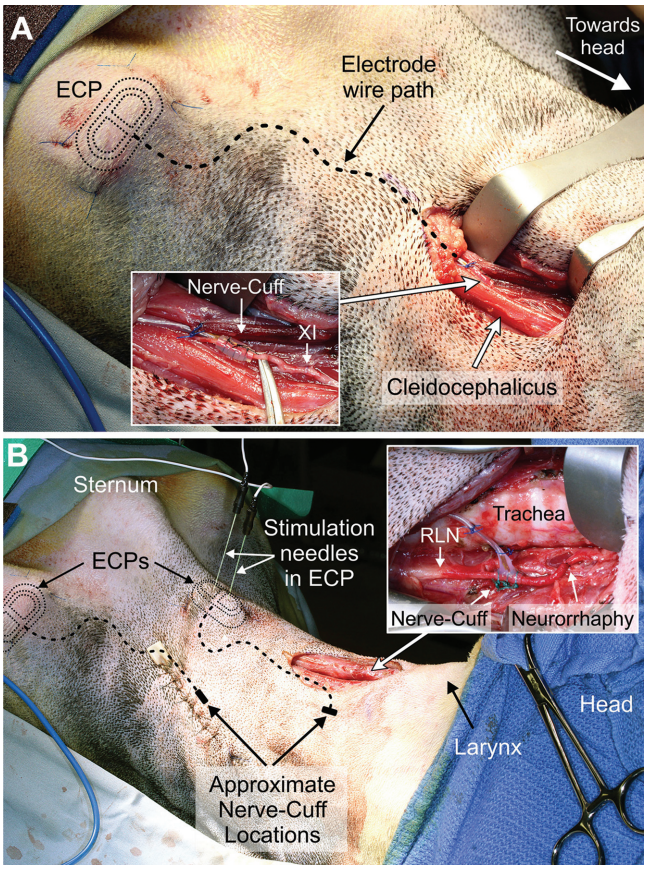


Figure 4. Nerve cuff electrodes with electrical connection ports (ECPs) are implanted on (A) the spinal accessory nerve and shoulder and (B) then the recurrent laryngeal nerve (RLN) and neck midline, respectively, in preparation for chronic testing. © Voice Health Institute. Used with permission of the Voice Health Institute.

surface, and needle advancement stopped when the tip reached the solid ECP floor. Before needle insertion, the skin surface above ECPs was clipped of fur and sanitized (ChloraPrep; BD Inc). Nerve regeneration was quantified by stimulating CMAPs, which were detected via hooked wire electromyography (EMG) using recording electrode pairs (40 AWG, 316 stainless steel with polymer jacket stripped 5-6 mm; AS765-40 [Cooner Wire]) inserted into the body of each vocal fold (eg, the thyroarytenoid and lateral cricoarytenoid complex [TA/LCA]) and posterior cricoarytenoid (PCA) muscle. Hooked wires were inserted transorally with a 23G carrier needle on an 8.25-in cannula (Pilling 507453; Teleflex). Electrode signals were band pass filtered (10-2000 Hz) and digitized (20 kS/s) with Axon Instruments hardware (Cyberamp 3800, Digidata 1440A) and software (Axoscope 10.4; Molecular Devices).

Contraction of the CC was elicited by stimulation through spinal accessory nerve cuffs, and TA muscle contraction was elicited by stimulation through intramuscular electrodes. Contractions were qualitatively judged by visual observation of initial twitch thresholds of neck movements and vocal fold adduction. Stimulation was delivered via a pulse generator (model S88; Grass Technologies Inc) and stimulus isolation unit (STMISOLA; Biopac Systems, Inc). Pulse trains were monophasic square waves that ranged from 1 Hz (to elicit CMAPs) to 80 Hz (to elicit fused contraction) across sub- to suprathreshold amplitudes (0.01-3 mA).

Assessment of ECP Tissue Encapsulation

At the end of experimental survival periods, ECPs were exposed through an incision and photographed to document local tissue encapsulation or other reaction. Photographs were also taken of explanted ECPs and electrodes and compared with preimplantation photographs to note any potential changes in surface characteristics. Tissues surrounding 3 of the ECPs fabricated from 3-dimensionally printed photopolymer were fixed in 10% formalin, paraffin embedded, sectioned at 5 μm, and stained with standard protocols for hematoxylin and eosin and Masson trichrome.

Results

Bench-Top ECP Testing

The direct current resistance and alternating current impedance of ECP/electrode electrical connections were assessed prior to implantation. The resistance added by ECPs was 2.14 ± 0.9 Ω (mean ± SD), which was negligible given the 200- to 2000-times greater resistance experienced at the typical electrode-tissue interface (approximately 0.5-5 kΩ). Impedance (at 1 kHz) was <100 Ω (below the measurable scale). The potential current leak from an ECP submerged in saline solution did not increase across 200 needle penetrations through the ECP's silicone surface, with impedance remaining >4.6 MΩ.

Implanted Electrode Testing Through ECPs

TA Stimulation With Intramuscular Electrodes. In 2 dogs, 3 bipolar intramuscular electrodes were implanted in the TA muscle (see Methods). The electrode impedance (at 1 kHz) at the time of implantation was approximately 500 Ω and remained <700 Ω for the up-to-22-week survival periods. These animals had elevated TA stimulation thresholds due to laryngeal nerve injury (as part of a different study[6,8]), but vocal fold adduction was observed with pulse trains of 40-Hz, 1-mA square waves with 1-millisecond pulse width.

Cleidocephalicus Contraction With Spinal Accessory Nerve Cuff Electrodes. Eight cuff electrodes were implanted onto the spinal accessory nerve supplying the CC muscle in 6 dogs. These electrodes were used to elicit contraction of the CC as a trigger for closed-loop laryngeal functional electrical stimulation testing done in the context of a separate study.[6,8] These implants were tested 39 times between implantation and either animal euthanasia or survival and surgical removal of the implant (**Figure 5**). Nerve cuff stimulation through ECPs at the time of implantation had CC twitch threshold amplitudes of <0.1 mA (pulse width ≤0.3 milliseconds) for all 8 implants. For 3 implants, the spinal accessory nerve had dislodged from the cuff electrode

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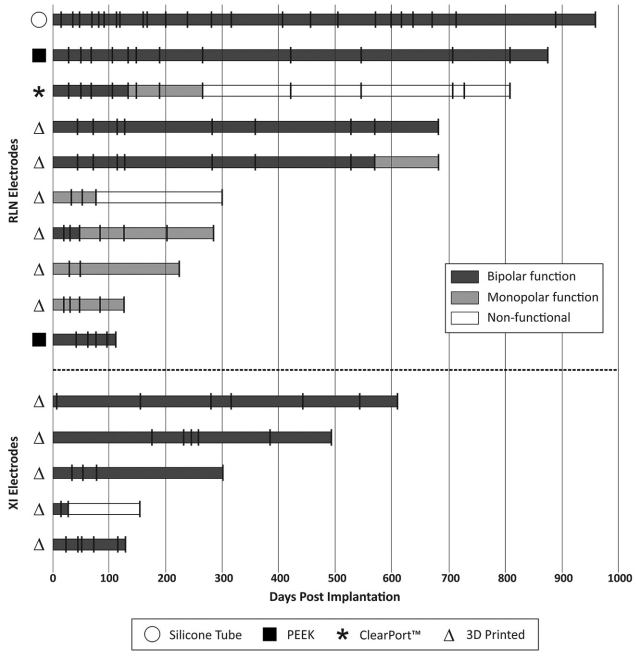


Figure 5. Testing time points (vertical marks) for 10 recurrent laryngeal nerves (RLNs) and 5 XI nerve cuff electrodes attached to 4 electrical connection port designs. Functional status for each implant is indicated by horizontal bar shading.

within the first month, precluding effective CC stimulation. These 3 cuffs are excluded from **Figure 5**. One cuff on the spinal accessory nerve initially elicited low-threshold CC contraction (<0.1 mA) at days 14 and 27 but was subsequently removed due to infection. The remaining 4 cuffs on spinal accessory nerves were working in a bipolar manner at their latest testing days of 128, 301, 493, and 610. The 3 cuffs implanted for >300 days all had relatively low CC twitch thresholds, ranging between 0.04 and 0.15 mA (0.3-millisecond pulse width).

Intrinsic Laryngeal Muscle Stimulation With RLN Cuff Electrodes. In 7 dogs, 11 bipolar nerve cuff electrodes were placed on 1 or both RLNs proximal to transection and epineurial suture repair to track laryngeal reinnervation (as part of a separate study[6,8]) across a total of 105 postimplantation recording sessions (mean, 9.5 sessions/animal; see **Figure 5**). Nerve cuff electrodes with ECPs remained in place a mean 470 days with a range of 97 to 960 days. One dog had no spontaneous or stimulable laryngeal EMG/CMAPs or movements at the terminal time point of 97 days and was therefore excluded from **Figure 5** due to uncertainty regarding cuff failure versus nerve regeneration failure. Two electrodes failed after days 77 and 266 due to wires breaking between the ECP and electrode contacts, and 1 was removed prematurely due to infection at day 140. The remaining 7 electrodes were functional for at least 1 of the 2 electrical contacts until planned terminal experiments (mean, 546 days; range, 112-960 days).

Stimulation thresholds for vocal fold movement across the postsurgical period depended on the time course of each regenerating RLN. Variability in postsurgical recording schedules precluded a comprehensive evaluation of the RLN reinnervation process across consistent time points. In general, stimulable movement and CMAPs were noted as early as day 30 after RLN transection and suture repair, and stimulation thresholds in some instances returned to within the range of prelesion values (eg, <0.1 mA, 0.3-millisecond pulse width). In 9 of the 11 nerve-cuffed RLNs, multiple recordings in each animal were able to capture an initial appearance of spontaneous EMG motor units on inspiration within the PCA, followed at later time points by synkinetic inspiratory units appearing in the TA/LCA as well. RLN stimulation in these animals likewise showed initially stronger PCA CMAPs than TA/LCA CMAPs and elicited vocal fold abduction, followed at later time points by progressively stronger elicited adduction that eventually overpowered abduction.

A sample time course of RLN recovery is shown after bilateral RLN transection and suture repair in **Figure 6**. In this animal, EMG was absent after nerve injury (not shown) and began to reappear in the PCAs by day 44. Activation of motor units in the vocal fold TA/LCA complex developed later than PCA reinnervation and represents synkinesis (misrouted reinnervation) because EMG activity is normally present in the PCA only during inspiration in propofol-anesthetized dogs, as evident in the preinjury recording. CMAPs elicited by maximal-response amplitude RLN stimulations (at the dotted line in **Figure 6**) were substantially larger and shorter in latency at the preinjury time point as compared with CMAPs during reinnervation. The regenerating CMAPs were shorter in latency for the PCA than TA/LCA and generally got larger, less polyphasic, and shorter in latency across time points as reinnervation matured (note y-axis scale changes across time in **Figure 6**).

ECP Tissue Encapsulation. The foreign body response to ECPs and electrode materials was a relatively thin capsule (~100-400 μm thick), suggesting that the materials used in ECP fabrication elicited a similar response as implant-grade silicone (capping the ECP compartments) and that the conductive filler within ECP compartments was adequately contained or elicited little tissue reaction. Trichrome and hematoxylin and eosin staining indicated that the capsule resembled scar tissue and was mainly composed of coarse collagenous fibers, fibroblasts and some blood vessels (**Figure 7**).

Discussion

Subcutaneous ports have long been used for periodic hypodermic needle access to cannulated blood vessels for drug delivery and blood sampling, and the same concept can be used to create ECPs for periodic needle electrode connection with chronically implanted nerve cuff and intramuscular

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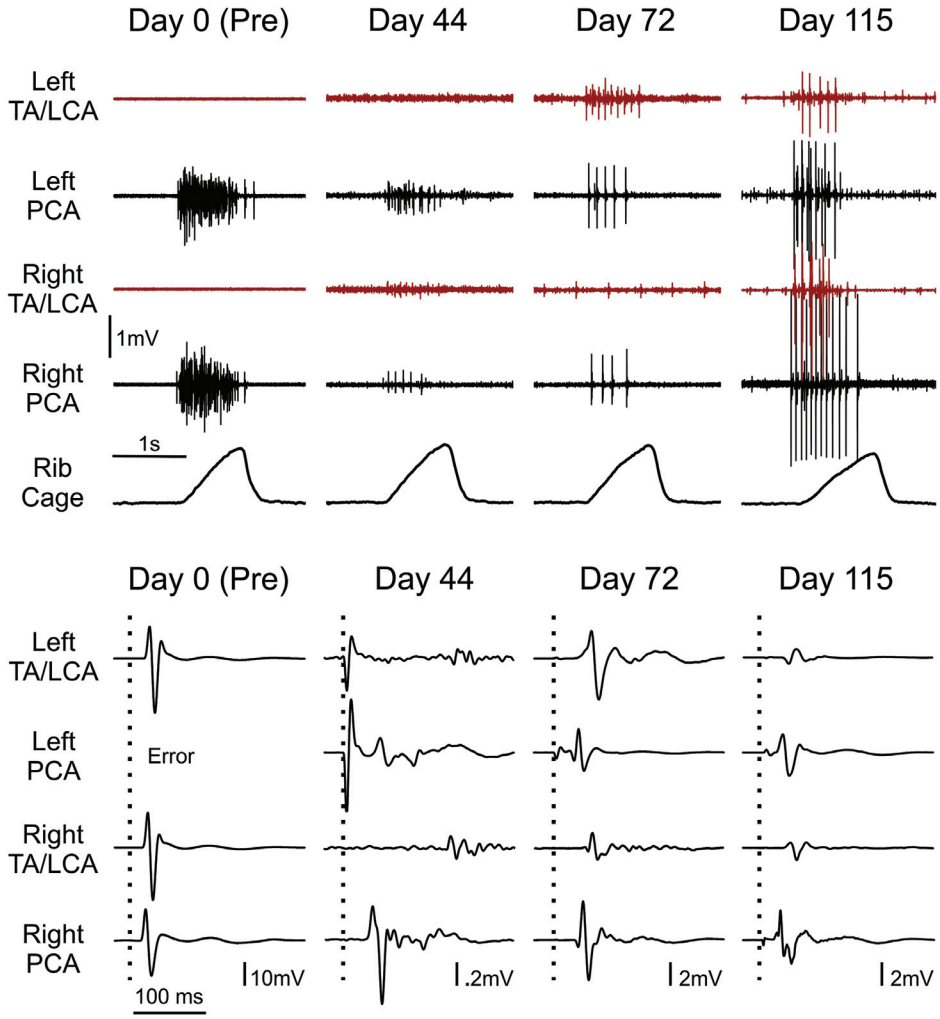


Figure 6. Spontaneous electromyography during inspiration (top) and elicited compound muscle action potentials (bottom) for a dog before bilateral recurrent laryngeal nerve transection and suture repair (day 0) and at 3 time points after (days 44, 72, 115). LCA, lateral cricoarytenoid; PCA, posterior cricoarytenoid; TA, thyroarytenoid.

electrodes in animal models.[3-5] This report describes our development and testing of multiple ECP design iterations, initially adapting readily accessible materials (eg, silicone tubing and polymer fittings) and ultimately printing ECP shapes with medical-grade photopolymer. Although wireless telemetric connections with chronic implants are available in several commercial systems, these devices are relatively expensive and are usually constrained in function (eg, have limited stimulation capabilities or a specific sensor type). Intermittent needle connection with ECPs enabled us to track the progress of laryngeal reinnervation after RLN transection and suture repair, observing the progression from initial PCA reinnervation (and stimulated glottic abduction) to the eventual reinnervation of TA/LCA musculature (and stimulated adduction) consistent with intralaryngeal RLN branching patterns.[9] In addition, ECPs were reliable over years of implantation, with instances of electrode failure attributable to fragile electrode elements instead of ECPs themselves.

The original concept for ''data ports'' was proposed for accessing implanted sensors[4,5] and recording EMG signals from bare electrode terminals.[3] Although ECPs were used only for delivering stimulating currents in the present study, ECPs should also be capable of accessing laryngeal EMG activity over time from intramuscular electrodes without the need for repeated needle insertion into the small laryngeal muscles. We have used chronic intramuscular electrodes like those in the present study to obtain spontaneous laryngeal EMG in awake, active canine subjects using a wireless EMG system[6] (Ripple Neuro Inc), and future studies are needed to determine whether ECPs can be used to obtain EMG with similar fidelity as commercial telemetric systems or direct percutaneous connections.[10,11]

In this study, >300 individual needle electrode insertions were made through the skin into ECPs during periodic electrode testing, spanning years in some animals, without indications of infection or lasting tissue harm at the insertion sites. Two electrodes were removed due to infection; yet,

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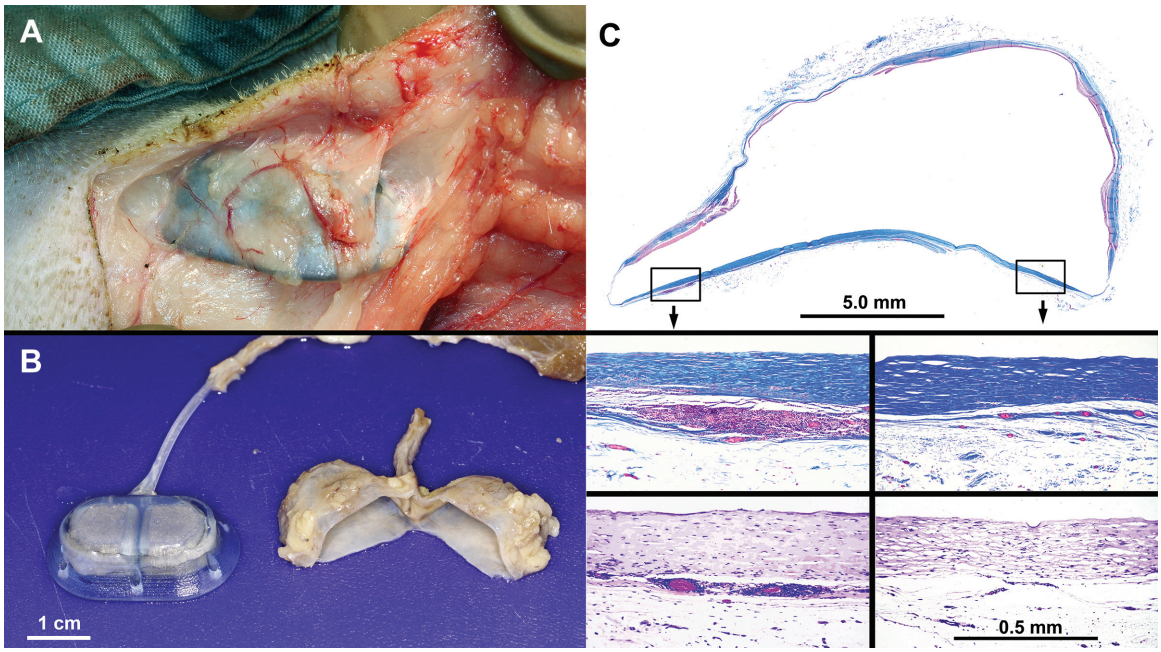


Figure 7. (A) Electrical connection port is explanted at day 455, with (B) the tissue capsule removed and opened up after fixation and (C) histologic staining with Masson trichrome (top and middle panels) and hematoxylin and eosin (bottom panels).

signs of infection were present in these animals before the initiation of postimplantation testing (needle insertions), so they likely stemmed from the implantation surgery. ECP materials elicited little reaction from tissue at implant locations, becoming encapsulated by a thin layer of connective tissue in a manner resembling encapsulation of silicone alone. Furthermore, the photopolymer used to print ECPs in this study did not appear altered after a year or more of implantation, and needle insertions did not appear to compromise the nonconductive compartment surface layer in testing, with hundreds of bench-top needle insertions or repeated in vivo insertions.

Although 7 implanted electrodes were stimulable through ECPs across survival periods lasting over a year in this study, several electrodes failed shortly after implantation. Revision surgery or postmortem examinations revealed that these electrode failures were attributable to wire breaks near stimulating contacts or dislodged cuff electrodes and that none of the electrode failures were due to a failure in the ECP aspect of the chronic implants. In the future, constructing implant electrodes with soft, stretchable conductive material instead of rigid wires may help prevent wire breaks or nerve cuff displacement. For example, intramuscular electrodes that use lines of platinum particle–impregnated silicone instead of conventional wires can withstand millions of flex cycles without losing conduction,[12] and many types of soft ''wires'' can stretch to double their resting length without breaking.[13] Future electrode designs could also test whether the ECP channel count can be expanded to support multichannel arrays and whether the overall size can be reduced while still providing reliable needle placement into the port compartments.

Conclusion

ECPs enable direct communication with chronically implanted electrodes via temporary needle insertion for long-term physiologic studies and preclinical device testing and development in animal models. Custom ECP designs are easily 3-dimensionally printed with biocompatible photopolymers that cause little tissue reaction over months or years of subcutaneous implantation. Future work should explore ECP designs capable of providing multiple reliable electrical connections while minimizing port size and reducing points of electrode failure.

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Author Contributions

James T. Heaton, co-conceived the project, assisted in implant design and fabrication, assisted in data acquisition through animal surgery and testing, analyzed the data, and cowrote the manuscript; **James B. Kobler**, co-conceived the project, assisted in data acquisition through animal testing and cowrote the manuscript; **Mark P. Ottensmeyer**, designed and fabricated implant devices and revised the manuscript; **Robert H. Petrillo**, assisted in implant fabrication, assisted in data acquisition through animal testing and revised the manuscript; **Monica A. Tynan**, assisted in data acquisition through animal testing and revised the manuscript; **Robert E. Hillman**, co-conceived the project, assisted in data acquisition through animal testing, and revised the manuscript; **Steven M. Zeitels**, co-conceived the project, assisted in data acquisition through animal surgery, and revised the manuscript.

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Disclosures

Competing interests: None.

Sponsorships: None.

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